TIDYWELLS REFERENCE SHEET

Decline Curves, One Page

The Arps family in the order you will meet it.

Exponential (b = 0)

$$q = q_i \cdot e^{-Dt}$$

A constant share of the rate is lost each period. Often used as a floor for late life.

$$N_p = (q_i - q) / D$$

Cumulative production to rate q.

Hyperbolic (0 < b < 2)

$$q = q_i / (1 + b \cdot D_i \cdot t)^{1/b}$$

The decline rate itself falls as the well ages. Most unconventional wells need b above 0.

Harmonic (b = 1)

$$q = q_i / (1 + D_i \cdot t)$$

A special case of the hyperbolic family. Cumulative grows without bound, so it needs a cutoff.

Effective vs nominal

$$D_e = 1 - e^{-D} \quad \text{(exponential)}$$

Nominal D is the continuous rate; effective D_e is the share lost over a year. Mixing them up is the most common error.

What the symbols mean

SYMBOL	MEANING
q	rate at time t
q_i	initial rate at the start of the decline
D_i	initial nominal decline rate (per unit time)
b	decline exponent (curvature)
t	time since the start of the decline

Habits that keep a fit honest

- Fit from the peak rate, not from first production.
- Convert monthly volumes to daily rates first.
- Bound b (Tidywells uses 0 to 2) so a noisy tail cannot produce nonsense.
- Report the range, not one number. A fit is a description of the past, not a promise.
- Shut-ins, workovers and curtailments break the curve. Mark them and refit.

Reference: Arps, J.J. (1945), Analysis of Decline Curves, Transactions of the AIME. Illustrative only; not for reserves reporting or investment decisions.